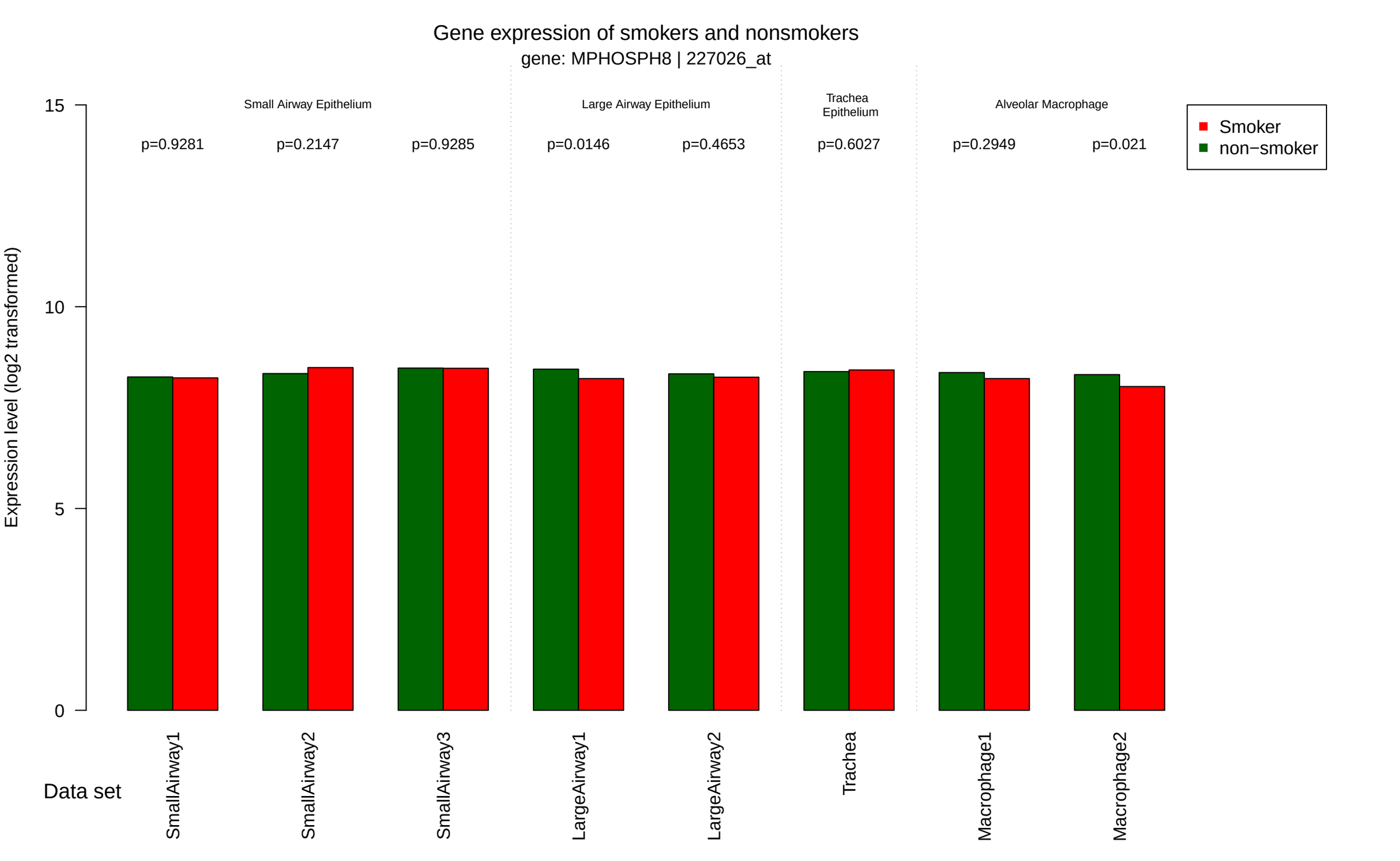
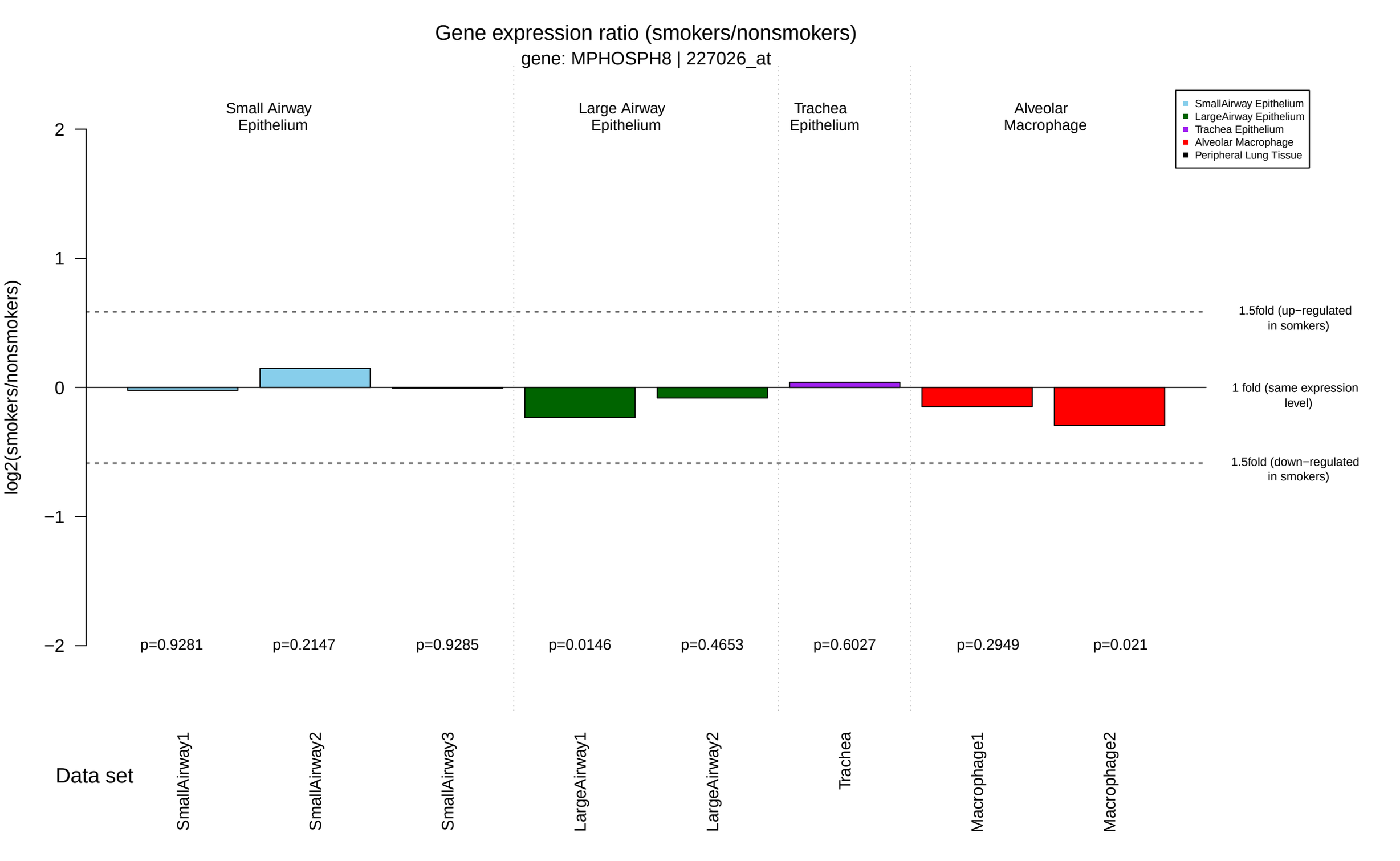
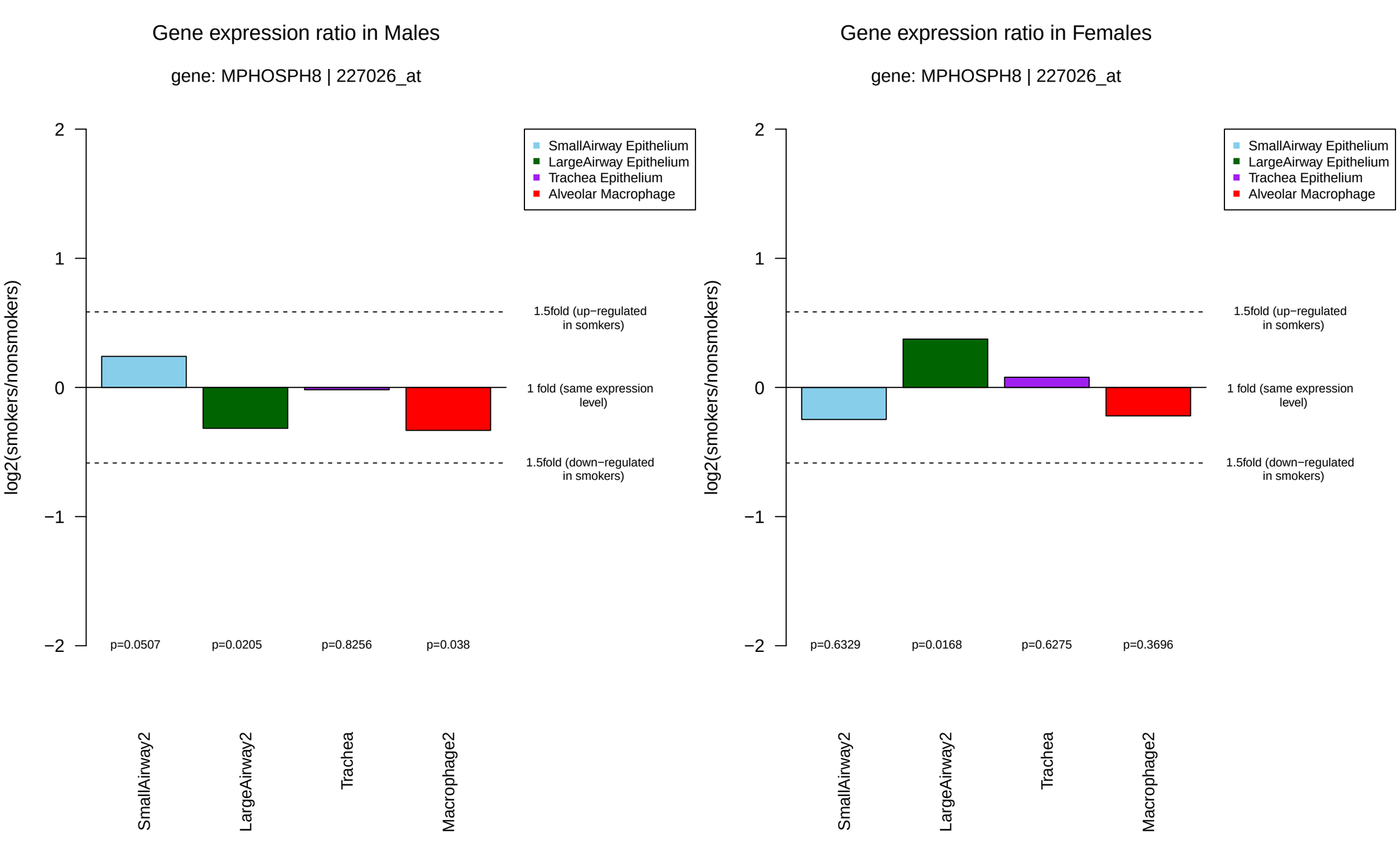
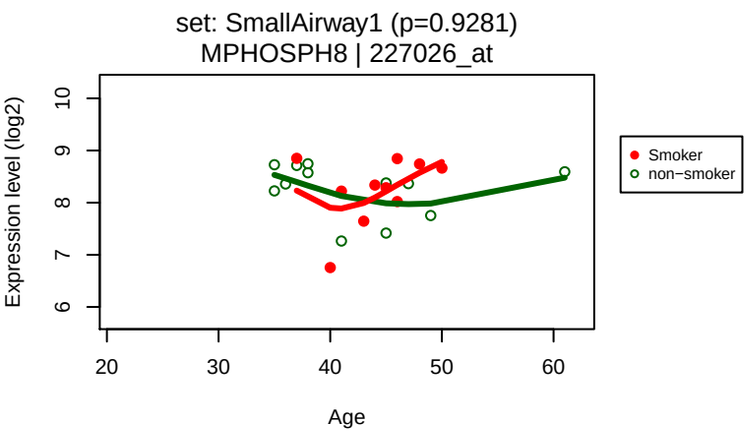
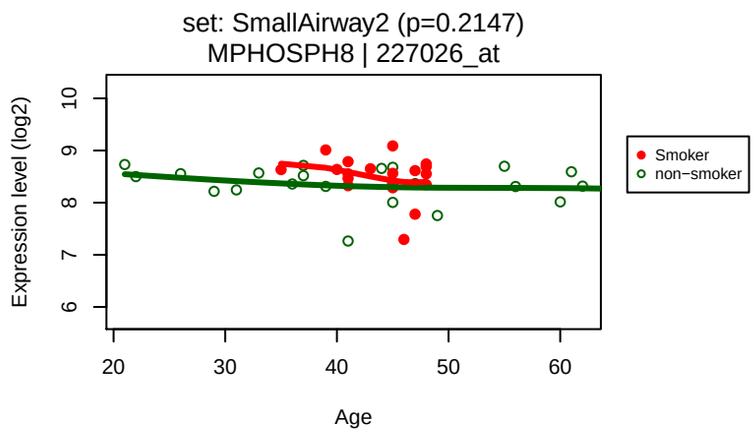
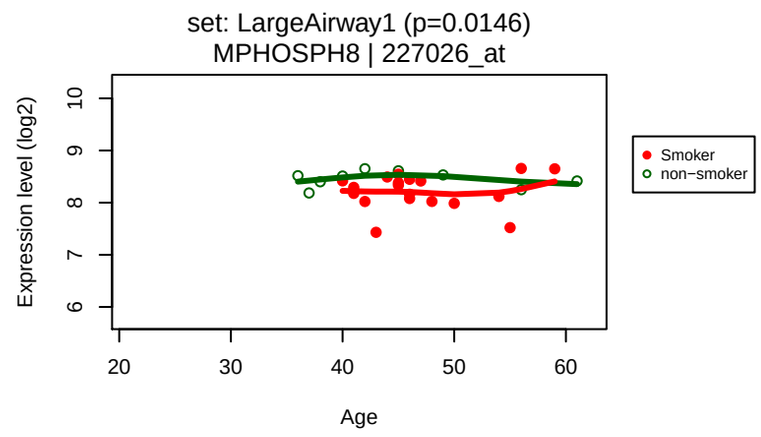

Gene expression of smokers and nonsmokers
gene: MPHOSPH8 | 227026_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.9281
p=0.2147
p=0.9285
p=0.0146
p=0.4653
p=0.6027
p=0.2949
p=0.021
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

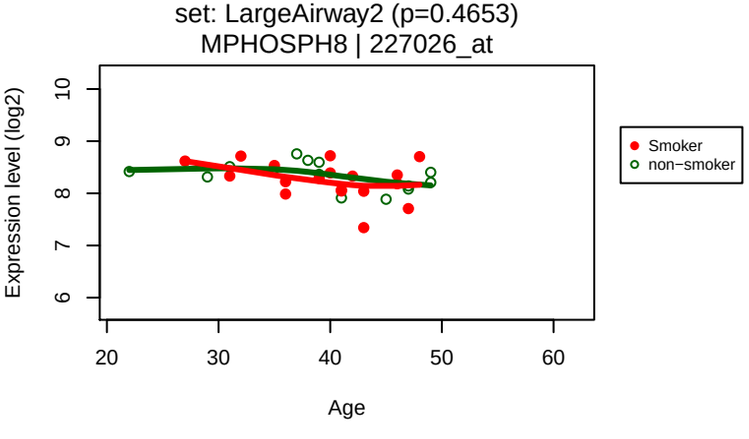

Gene expression ratio (smokers/nonsmokers)
gene: MPHOSPH8 | 227026_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9281
p=0.2147
p=0.9285
p=0.0146
p=0.4653
p=0.6027
p=0.2949
p=0.021
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

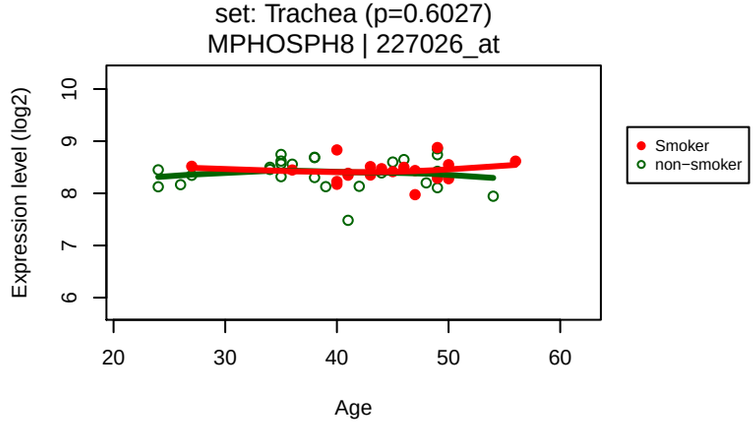

Gene expression ratio in Males
gene: MPHOSPH8 | 227026_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0507
p=0.0205
p=0.8256
p=0.038
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: MPHOSPH8 | 227026_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6329
p=0.0168
p=0.6275
p=0.3696
SmallAirway2
LargeAirway2
Trachea
Macrophage2

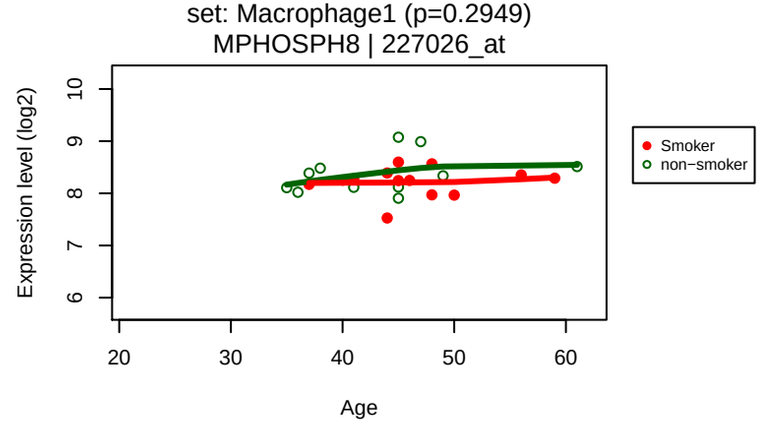

set: SmallAirway1 (p=0.9281)
MPHOSPH8 | 227026_at
Expression level (log2)
Age
Smoker
non−smoker

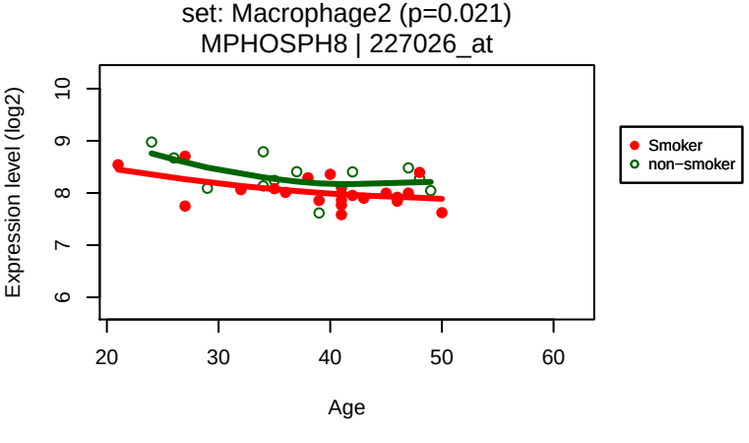

set: SmallAirway2 (p=0.2147)
MPHOSPH8 | 227026_at
Expression level (log2)
Age
Smoker
non−smoker

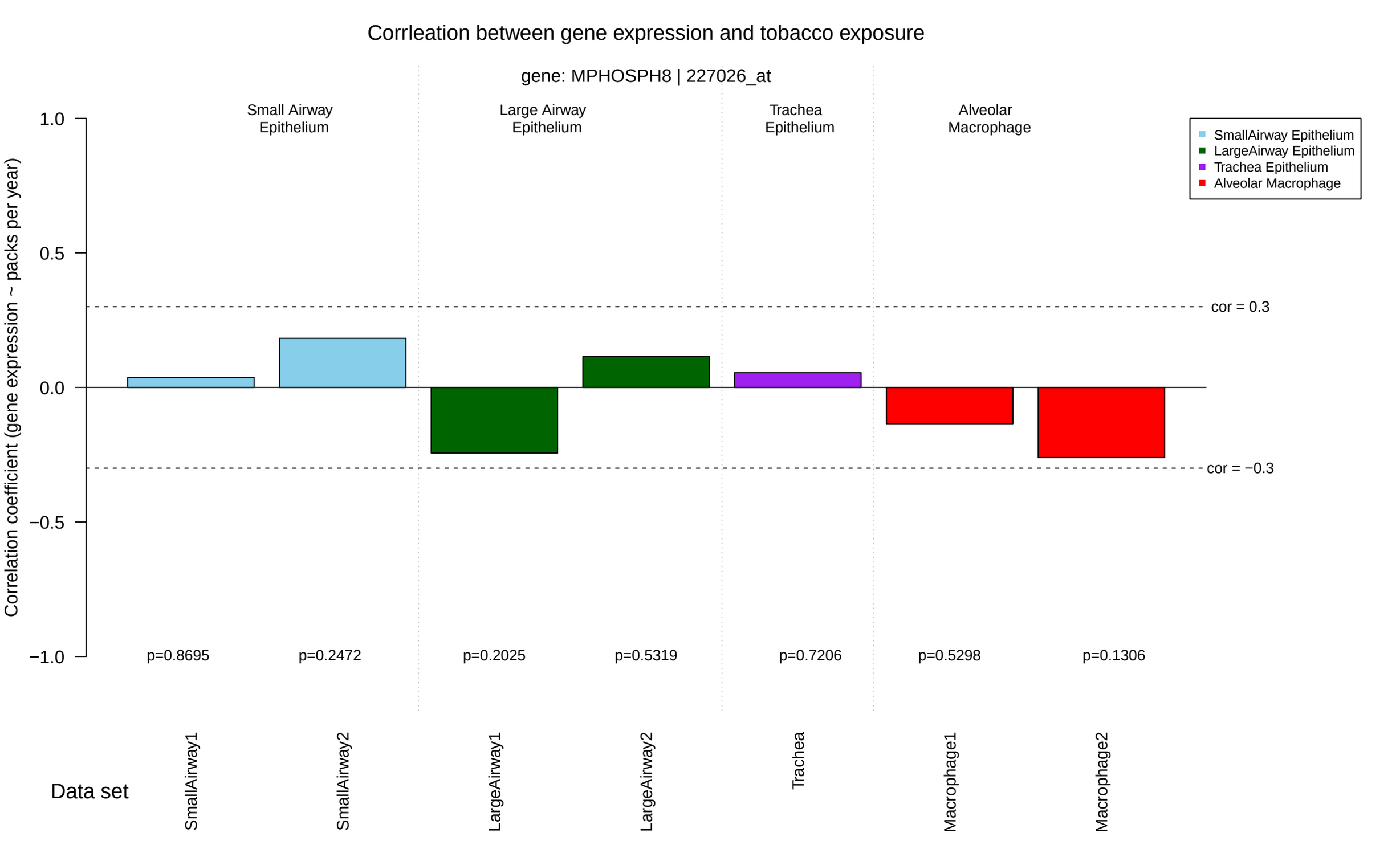

set: LargeAirway1 (p=0.0146)
MPHOSPH8 | 227026_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.4653)
MPHOSPH8 | 227026_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.6027)
MPHOSPH8 | 227026_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.2949)
MPHOSPH8 | 227026_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.021)
MPHOSPH8 | 227026_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: MPHOSPH8 | 227026_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.8695
p=0.2472
p=0.2025
p=0.5319
p=0.7206
p=0.5298
p=0.1306
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2